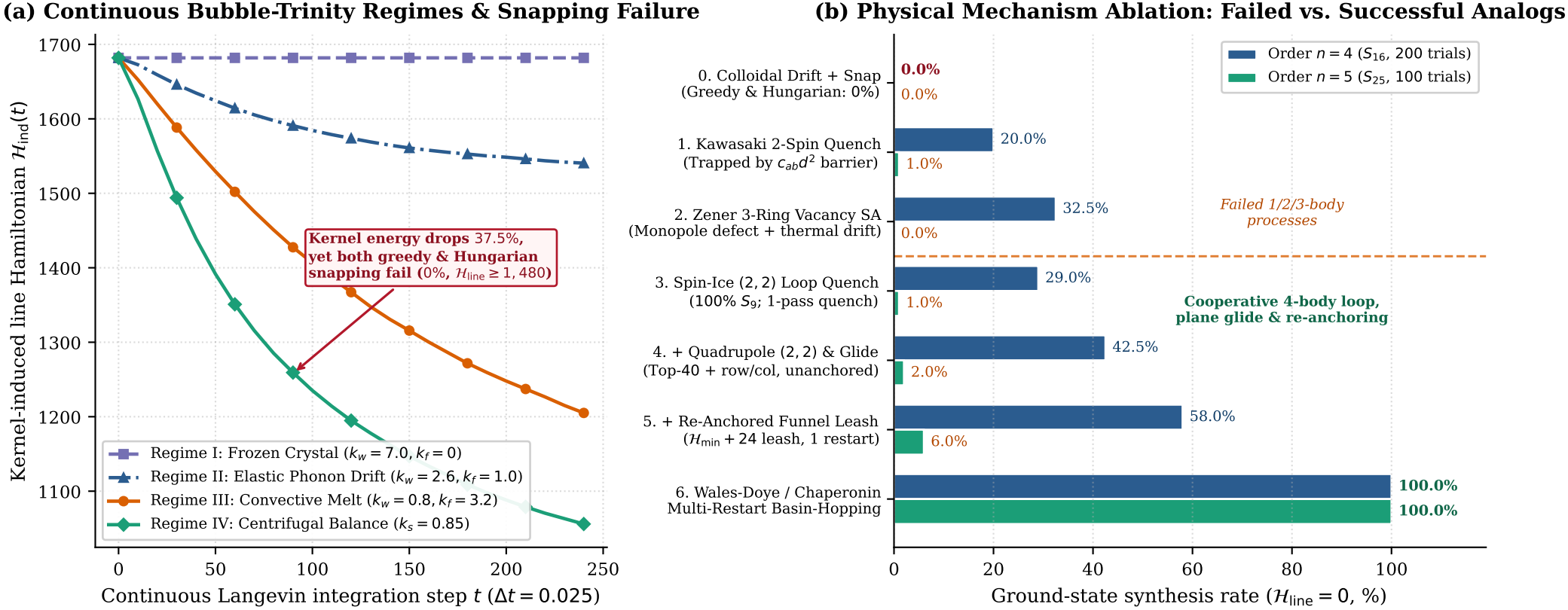

(a) Continuous Bubble-Trinity Regimes & Snapping Failure
Kernel-induced line Hamiltonian $\mathcal{H}_{\text{ind}}(t)$
Continuous Langevin integration step t ($\Delta t = 0.025$)
Kernel energy drops 37.5%, yet both greedy & Hungarian snapping fail (0%, $\mathcal{H}_{\text{line}} \geq 1,480$)
Regime I: Frozen Crystal ($k_w = 7.0, k_f = 0$)
Regime II: Elastic Phonon Drift ($k_w = 2.6, k_f = 1.0$)
Regime III: Convective Melt ($k_w = 0.8, k_f = 3.2$)
Regime IV: Centrifugal Balance ($k_s = 0.85$)
(b) Physical Mechanism Ablation: Failed vs. Successful Analogs
Order $n = 4$ (S_{16}, 200 trials)
Order $n = 5$ (S_{25}, 100 trials)
0. Colloidal Drift + Snap (Greedy & Hungarian: 0%)
1. Kawasaki 2-Spin Quench (Trapped by $c_{ab}d^2$ barrier)
2. Zener 3-Ring Vacancy SA (Monopole defect + thermal drift)
3. Spin-Ice (2, 2) Loop Quench (100% S_9; 1-pass quench)
4. + Quadrupole (2, 2) & Glide (Top-40 + row/col, unanchored)
5. + Re-Anchored Funnel Leash ($\mathcal{H}_{\text{min}} + 24$ leash, 1 restart)
6. Wales-Doye / Chaperonin Multi-Restart Basin-Hopping
0.0%
0.0%
20.0%
1.0%
32.5%
0.0%
29.0%
1.0%
42.5%
2.0%
58.0%
6.0%
100.0%
100.0%
Failed 1/2/3-body processes
Cooperative 4-body loop, plane glide & re-anchoring
Ground-state synthesis rate ($\mathcal{H}_{\text{line}} = 0$, %)